

Name: _____ Partner: _____ Period: _____

Data Dispatch Agency

Your team works at the Data Dispatch Agency. Four requests just arrived, each describing a real-world situation that needs a function model. For each dispatch, your team must (1) select the best model type, and (2) state the key assumptions and domain restriction. Work through the tier that matches your group's assignment.

The four dispatches:

- **Dispatch 1.** A city tracks its population each year. The population increases by approximately 850 people per year for the past decade.
- **Dispatch 2.** A farmer wants to fence a rectangular pen using 200 m of fencing. She records the area for different choices of one side length.
- **Dispatch 3.** A warehouse stores boxes stacked in cubic arrangements. The total number of boxes for a stack of side length n units is recorded.
- **Dispatch 4.** A ride-share app charges \$2.50 per mile for the first 5 miles, then \$1.75 per mile for each mile after that.

Tier R — Remediate

For each dispatch, **circle the best model type** from the three choices given, then complete the assumption sentence.

Dispatch	Best model (circle one)			Complete the assumption
1	Linear	Quadratic	Cubic	"This model assumes the population grows by a _____ amount each year."
2	Linear	Quadratic	Cubic	"This model assumes the pen is a _____ and the perimeter is exactly _____ m."
3	Linear	Quadratic	Cubic	"This model assumes the stacks form perfect _____ shapes."
4	Linear	Quadratic	Piecewise	"This model assumes the rate changes at exactly _____ miles."

Tier A — Apply

For each dispatch, compute differences from the table provided, determine the degree, name the model type, and state one assumption with domain restriction.

Dispatch 1 table:

Year (x)	Population (y)	Δy
0	12,000	—
1	12,850	_____
2	13,700	_____
3	14,550	_____

Model type: _____ Degree: _____

Assumption: _____

Domain restriction (give an inequality): _____

Dispatch 2 table: (side length ℓ in meters, area A in m^2)

ℓ	A	ΔA	$\Delta^2 A$
10	900	—	—
20	1600	_____	—
30	2100	_____	_____
40	2400	_____	_____
50	2500	_____	_____

Model type: _____ Degree: _____

Domain restriction: _____

Tier E — Extend

- Dispatch 4 — Piecewise model.** Write a two-piece description of the ride-share cost function $C(m)$ where m is the number of miles driven.

Piece 1 (applies when _____): $C(m) =$ _____

Function type for Piece 1: _____

Piece 2 (applies when _____): $C(m) =$ _____

Function type for Piece 2: _____

- State **two assumptions** embedded in the Dispatch 4 model and explain why each assumption is needed for the model to be valid.

Assumption 1: _____ Why it matters: _____

Assumption 2: _____ Why it matters: _____

- For Dispatch 3, a student claims a cubic model is appropriate because “volume is three-dimensional.” Explain in one or two sentences whether this reasoning is mathematically sound, referencing differences.

